2004 IEEE International Conference on Systems, Man and Cybernetics

Moving Blocks Bootstrap Control Chart for Dependent Multivariate Data *

Yu-ming Liu
National Lab of Industrial Control Technology
Zhejiang University
Hangzhou, 310027, P.R.China
ymliu@iipc.zju.edu.cn

Jun Liang
National Lab of Industrial Control Technology
Zhejiang University
Hangzhou, 310027, P.R.China
jliang@iipc.zju.edu.cn

Ji-xin Qian
National Lab of Industrial Control Technology
Zhejiang University
Hangzhou, 310027, P.R.China
jxqian@iipc.zju.edu.cn

Abstract - *A method for constructing moving blocks bootstrap control chart for dependent multivariate data was proposed. In this method the statistics of the control charts were firstly obtained by principal component analysis, then a modified MBB was used to determine their control limits which can not be based on independent and identically distributed(IID) assumption. Empirical average run length and false alarm rate were proposed to evaluate the control charts. Two examples, adopting simulation data and real industrial data respectively, were given to illustrate the method, in which some results about factors' impact on MBB, such as sample size , moving block size, block number et al, were obtained. The results show MBB-based method is superior to PCA-based method under certain conditions and is an available way to establish control charts for weakly dependent multivariate data.*

Keywords: moving blocks bootstrap, principal component analysis, control chart, dependent multivariate data, average run length, control limit

1 Introduction

The bootstrap is a powerful non-parametric technique for estimating the sampling distribution of a given statistic, first introduced by B. Efron(1979), which utilizes mass computation instead of traditional normal assumption, and is generally a better choice when the sampling distribution is unknown or has certain skewness or the size of samples is limited. The application of bootstrap to statistical process control(SPC) has given some attentions[3][4][8], and N. Choomrit (2001) concluded some application criteria for bootstrap-based control chart.

However, the traditional bootstrap can only be applied to univariate independent sample. For dependent multivariate data, R. Y. Liu and J. Tang (1996) illustrated that combining data depth approach(R. Y. Liu, 1995) and the moving blocks bootstrap(MBB) ideas may lead to a new nonparametric approach for statistical control of dependent multivariate observations ,but gave no details. Compared with data depth for multivariate data, principal component analysis(PCA) is a more popular technique for extracting the collinear relationship between the variables and reducing data dimension, and the principal component scores are easy to be interpreted either mathematically or geometrically. Furthermore, Hotelling's T^2 statistic and Q statistic can be constructed based on the retained principal components and the multivariate control chart then is transformed to an univariate one. One of the difficulties of using PCA for dependent multivariate data monitoring is that the scores from PCA and the statistic T^2 or Q can't approximated by certain parametric distribution because the observations don't satisfy IID assumption. Therefore, accurate control chart is hard to develop. To address the problem, EWMA(exponentially weighted moving average) was used on the score data [2][7], and some nonparametric techniques ,such as kernel density estimate, were also investigated to estimate the corresponding statistics' distribution. But no general solution scheme has been achieved.

Thus, this paper proposed a method for applying moving blocks bootstrap(MBB) to construct control charts for dependent multivariate data, which is the common case in practice, i.e. in chemical process industries. The basic principle of this method is utilizing principal component analysis(PCA) to get the principal component scores T , T^2 statistic and Q statistic, then determine their control limits by MBB. During the establishment of MBB control charts for the statistics, some factors ,such as observations size N , number of resamples B, block size b , block number k et al, were taken into account. Factor analysis and extensive simulation experiments were suggested to determine the corresponding parameters. Empirical average run

* 0-7803-8566-7/04/$20.00 © 2004 IEEE.

Authorized licensed use limited to: UNIVERSITY OF NEW MEXICO. Downloaded on October 12,2026 at 16:31:59 UTC from IEEE Xplore. Restrictions apply.

length(ARL) and false alarm rate were proposed to evaluate the control limits. The procedure of the foregoing method was presented. Two examples were conducted to illustrate the MBB-based method. The data of first example is from multivariate normal distribution, the second one is from an industrial fluidized-bed reactor (FBR) under normal operation. The PCA-based method and MBB-base method are compared in the two examples. And the results show the MBB-based method is superior to PCA-based method under certain conditions. In addition, the influence of each factor mentioned above are concluded in the experiment results.

The remained sections of the paper is arranged as follows. In section 2, both basic concept of MBB control charts and two evaluation rules are presented, as well as the procedure of the MBB-based method. Section 3 shows two examples to illustrate the foregoing method. Conclusions are drawn in section 4.

2 MBB control chart

2.1 Construct statistics by PCA

Principal component analysis(PCA) is a multivariate statistical technique for data dimensionality reduction and features extraction, which is widely used in multivariate statistical process control(MSPC). Consider a n-dimensional data matrix $X_{PCA} \in R^{m_P \times m}$, where m_P is the number independent samples. The PCA decomposition of X_{PCA} can be defined as formula(1) :

$$X_{PCA} = TP^T + E = T_1P_1^T + T_2P_2^T + \ldots + T_{Nr}P_{Nr}^T + E_p \qquad (1)$$

Where Nr is the number of retained PCs; $T=(T_1,T_2,...,T_{Nr})$ is the matrix of principal component scores vectors; $P=(P_1,P_2,...,P_{Nr})$ is the matrix of principal component loading vectors; E_p is the residual matrix. For establishing PCA-based control charts, after the normal reference data are modeled by PCA representation, some statistics can be calculated and the corresponding warning or action limits can be defined. Once a statistic moves outside the control limits, it implies some abnormal operation may occur. Elements of scores vector T is the linear combination of the original data and can be monitored by conventional Shewhart control chart[6]. While Hotelling's T^2 statistic and Q statistic are generally adopted, the former is a measure of the Mahalanobis distance in the reduced space indicating the change of the data covariance, the latter defines the Euclidean distance of the position of an observation from the hyper plane formed by the PCA representation reflecting the change of the relationships between variables. The two statistics at sample time j can be defined as formula(2) and formula(3) respectively:

$$T_j^2 = \sum t_{ji}^2/\lambda_i \qquad (2)$$

$$Q_j = \sum\left(X_{jk} - \hat{X}_{jk}\right)^2 \qquad (3)$$

Where T_{ji} is the jth value of the ith retained principal component score vector; λ_i is the ith value of the covariate matrix.

Conventional PCA-based control limits assumes the observations follow independent and identically distribution, so the control limit of T^2 and Q for a significance level of α can be calculated by formula(4) and formula(5) respectively:

$$T_\alpha^2 = \frac{(N_s-1)(N_s+1)k}{N_s(N_s-k)}F_\alpha(k, N_s-k) \qquad (4)$$

$$Q_\alpha = \theta_1\left[\frac{C_\alpha\sqrt{2\theta_2h_0^2}}{\theta_1} + 1 + \frac{\theta_2h_0(h_0-1)}{\theta_1^2}\right]^{\frac{1}{h_0}} \qquad (5)$$

where N_s is the number of observations; k is number of the variables; $F_\alpha(k, N_s-k)$ is the upper $100\alpha\%$ critical point of the F distribution with k and N_s-k degrees of freedom.; $\theta_i = \sum_{j=k+1}^{n}\lambda_j^i$ with $i=1,2,3$, and $h_0 = 1-\frac{2\theta_1\theta_2}{3\theta_2^2}$; λ_j is the jth eigenvalue of the covariance matrix and C_α is the standard normal deviate corresponding to the upper $(1-\alpha)$ percentile.

2.2 Construct MBB control chart

The MBB generates the bootstrap samples from the moving blocks. Let $(S_1, S_2, \cdots S_N)$ be an original sample with size of N, its empirical distribution is F_N. The ith moving blocks of size b is defined as $B_i = (S_i, S_{i+1}, \cdots, S_{i+b-1})$, where $i=1,2,\cdots,N-b+1$, that is, the observations can be divided into $N-b+1$ blocks. It was presented in [4][6] that using the MBB with appropriate block size b, the observations belongs to different blocks are nearly independent while the correlation presents in observations of each block, implying observations in any blocks follow the same distribution. Draw k random blocks from the $N-b+1$ blocks with replacement for B times, form B MBB samples, each with size of kb, $S_i^* = (S_1^*, S_2^*, \cdots, S_{kb}^*)$, $i=1,2,\cdots,B$. Calculate the MBB

Authorized licensed use limited to: UNIVERSITY OF NEW MEXICO. Downloaded on October 12,2026 at 16:31:59 UTC from IEEE Xplore. Restrictions apply.

samples mean $\bar{S}_1^*, \bar{S}_2^*, \cdots, \bar{S}_B^*$, and define control limit with significant level α as $S_\alpha = (1-\alpha)$th empirical percentile.

The MBB method presented here is slightly different from the conventional MBB in [4][6]. [4] regarded MBB procedure as an extension of the standard bootstrap, and the MBB sample size kb equals the number of original observations N, that is $N = kb$. If the kb is too large, the smoothness of the observations will make a conservative guess for control limits in some cases, thus cause more false alarms. [4] also assumed $N = kb$ to obtain some propositions, but made a deduction that if b is large enough, then MBB sample formed by one block can approximate the k blocks MBB sample to obtain the distribution of $\bar{S}_1^*, \bar{S}_2^*, \cdots, \bar{S}_B^*$. However, in the MBB method proposed above, the k and b can be varied according to different cases, as long as $kb \le N$. It is apparent that the method in this paper is more flexible than the method in [4] and [6], and can be expected to gain a balance between smoothness of the data and the accuracy of the control limit by adjusting k and b.

It should be noted that the features of underlying distribution may play an important role on the accuracy of the MBB control charts. [4] did some research to discuss the applicability of subgroup bootstrap control charts under different skewed population, i.e. positive-skewed distributions, Gamma(0.5, 1); negative-skewed distributions, Weibull(20, 1), and concluded some results of the interacts among skewness , subgroup size and number of the observations. Since certain analogies exist between the subgroup bootstrap control chart and MBB control chart, it may be conjectured that skewness also has some effects on control charts constructed by MBB. Deeper research are required to provide some useful guidelines to take skewness into account, but it may be beneficial to check the sampling distribution according to the results of subgroup bootstrap before carrying out MBB.

2.3 Two Methods to evaluate control limit

2.3.1 Empirical ARL for simulation data

ARL is the average number of observations until a control chart identifies an observation outside the control limits. The in-control ARL can be calculated by formula(6) :

$$\mathrm{ARL} = \frac{1}{\alpha} \tag{6}$$

Where α is the false alarm rate or the Type I error probability. N_c simulation experiments are accomplished and N_c sets of control limits can be established either by MBB-based method or PCA-based method. Each control limit from the modeling N samples with confidence $(1-\alpha)$ is defined as $S_\alpha(i)$ and large number of testing samples , i.e. N_t samples, are used to calculate its empirical $\mathrm{ARL}(i)$ by formula(7) :

$$\mathrm{ARL}(i) = \frac{N_t}{N_b} \tag{7}$$

where N_b is the number of samples outside the control limit $S_\alpha(i)$. For N_c sets of control limits for the data from the same distribution, the corresponding ARLs refer to $\mathrm{ARL}(i)$, $i = 1, 2, \cdots, N_c$. The mean and standard deviation of $\mathrm{ARL}(i)$, $E(\mathrm{ARL})$ and $Std(\mathrm{ARL})$, are calculated by formula(8) and formula (9) respectively:

$$E(\mathrm{ARL}) = \frac{1}{N_c}\sum_{i=1}^{N_c}\mathrm{ARL}(i) \tag{8}$$

$$Std(\mathrm{ARL}) = \sqrt{\frac{1}{N_c - 1}\sum_{i=1}^{N_c}\left(\mathrm{ARL}(i) - E(\mathrm{ARL})\right)^2} \tag{9}$$

The control limits by MBB or conventional method are then evaluated by the $E(\mathrm{ARL})$ and $Std(\mathrm{ARL})$.

2.3.2 False alarm rate for real data

In relation to real data, it is unpractical to do large number of simulation experiments. In general, the control limits are constructed based on the representative real data under normal operation, then are evaluated by another testing data under the same conditions. The false alarm rate can act as evaluation rule. For testing data with limited size, the number of observations beyond the control limits approximately reflects the false alarm rate.

2.4 Procedure to construct MBB control chart

(1) Carry out autocorrelation function test and normality test to check the basic feature of the sampling data ;

(2) For observations with size of N, PCA is adopted to obtain principal component scores $\boldsymbol{T}$,statistic T^2 and Q, and their approximate distribution is analyzed by histogram;

(3) Divide the N observations into $N-b+1$ moving blocks with size of b ;

Authorized licensed use limited to: UNIVERSITY OF NEW MEXICO. Downloaded on October 12,2026 at 16:31:59 UTC from IEEE Xplore. Restrictions apply.

(4) Draw k random blocks from the $N-b+1$ blocks with replacement, form an MBB sample of size kb;

(5) Calculate the MBB sample mean $\bar{S}^*$;

(6) Repeat the steps(3)-(5) a large number of times, i.e. B;

(7) Sort the B MBB sample means $\left(\bar{S}_1^*, \bar{S}_2^*, \cdots, \bar{S}_B^*\right)$;

(8) For the desired false rate α, let S_α be the $100(1-\alpha)$ empirical percentile such that $(1-\alpha)B$ smallest ordered $\bar{S}^*$ below it. Then S_α is the $100(1-\alpha)\%$ control limit;

(9) The control limits are evaluated either by empirical ARL or false alarm rate α using testing data following the same distribution as the modeling data. If the control charts do not satisfied the requirement, repeat the foregoing steps.

3 Two examples of MBB control chart

3.1 For simulation data

Obtain 2002 observations from the multivariate normal distribution $N_3(\mu, \Sigma)$, where

$$\mu = [0\ 0\ 0],\ \Sigma = \begin{bmatrix} 1 & 0.95 & 0.9 \\ 0.95 & 1 & 0.85 \\ 0.9 & 0.85 & 1 \end{bmatrix}$$

Let $Y = (Y_1\ \ Y_2 \cdots Y_{2002})^T \in R^{2002\times 3}$, $Y_i \in R^{1\times 3}$, $i = 1,2,\cdots 2002$, and $X(k) = (Y_1\ \ Y_2 \cdots Y_{2000})^T \in R^{2000\times 3}$, $X(k+1) = (Y_2\ \ Y_3 \cdots Y_{2001})^T \in R^{2000\times 3}$. Next, another sample of size 2000 from $N(0,0.2)$ was generated as the measurement noise, refers to $E \in R^{2000\times 3}$. Then the 1-dependent multivariate simulation data are generated by formula(10):

$$X = X(k)A_1 + X(k+1)A_2 + E \qquad (10)$$

where

$$X = (X_1, X_2, \cdots, X_{2000})^T \in R^{2000\times 3}$$

$$X_i \in R^{1\times 3}, i = 1,2,\cdots 2000$$

$$A_1 = \begin{bmatrix} -3 & 9 & 10 \\ -1 & 16 & 6 \\ -7 & 10 & 4 \end{bmatrix},\ A_2 = \begin{bmatrix} 17 & 10 & 15 \\ 6 & 60 & 4 \\ -5 & 20 & -3 \end{bmatrix}$$

Let $X_m = (X_1, X_2, \cdots, X_N)^T$ as modeling data matrix, $X_t = (X_{1001}, X_{1002}, \cdots, X_{2000})^T$ as testing data. Autocorrelation test were carried out. Figure 1 shows the autocorrelation function of the first variable, the second principal component score and the T^2, with he sample size of $N = 50$. In this case, Q showed almost no autocorrelation, but other variable or statistics show dependent feature.

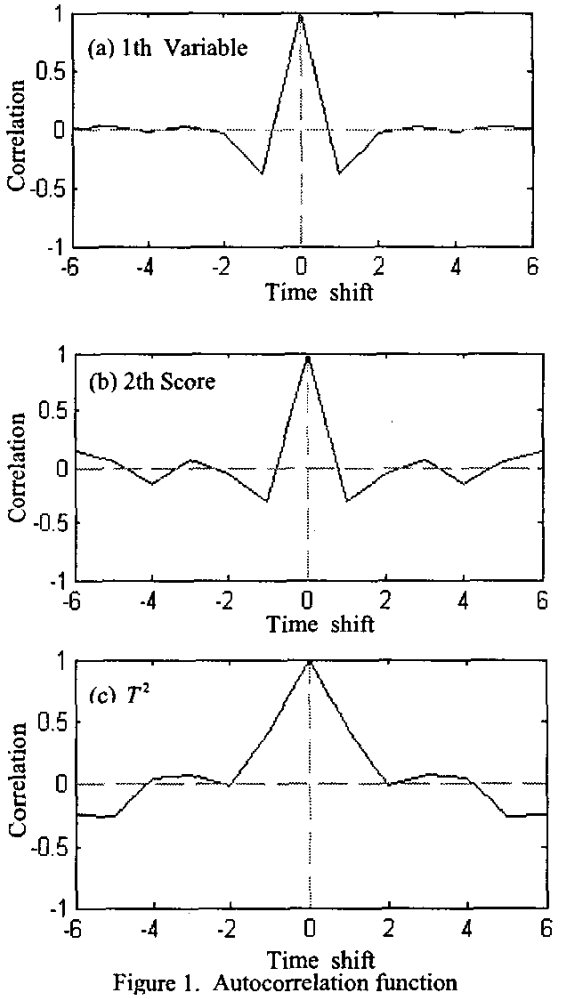


Figure 1. Autocorrelation function

To choose proper parameters to conduct MBB, different N, B, b and k are checked as shown in Figure 2. It can be seen that when B is large enough, then it has no impact on the control limits. B is relevant to the N, the bigger the N, the larger the B required, for example when $N = 100$, B should be more than 1000 to achieve the nearly steady control limits, while when $N = 50$, only

Authorized licensed use limited to: UNIVERSITY OF NEW MEXICO. Downloaded on October 12,2026 at 16:31:59 UTC from IEEE Xplore. Restrictions apply.

requires $B > 600$ as shown in Figure 2. (a) and (b) . As shown in (c) and (d), the control limits decrease as b or k increases

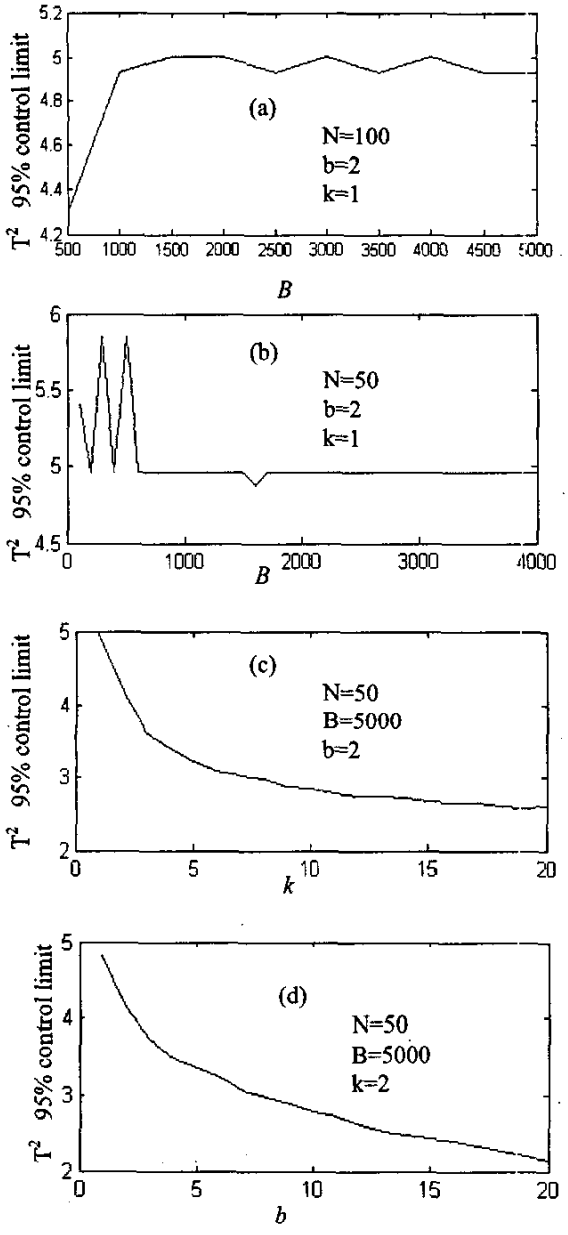


Figure 2. Parameters analysis

For different N, such as N=20, 50, 100, 200, 1000, different simulation experiments were conducted . Q and T^2 control charts were evaluated, while principal component scores charts are not drawn for consistency of comparison throughout this simulation. We found MBB-based method shows its effectives both in the case of large sample and small sample. Particularly, when N is small, MBB-based method has the potential to establish more accurate control limits than the PCA-based method. Table 1 shows their comparison based on empirical ARL, where $N_c = 100$ simulation experiments were conducted , N=20.

Table 1. Comparison of PCA-based method and MBB-based method control chart with 95% control limit

Sample size		20		
Block size b		1	2	2
Block number k		1	1	2
Q	$E(\mathrm{ARL})_{PCA}$	13.03	12.44	11.276
	$E(\mathrm{ARL})_{MBB}$	25.251	10.282	4.5671
	$Std(\mathrm{ARL})_{PCA}$	9.4867	9.3266	6.9518
	$Std(\mathrm{ARL})_{MBB}$	99.972	18.148	1.877
T^2	$E(\mathrm{ARL})_{PCA}$	Infinite	Infinite	52.811
	$E(\mathrm{ARL})_{MBB}$	27.58	19.518	6.6169
	$Std(\mathrm{ARL})_{PCA}$	----	----	105.02
	$Std(\mathrm{ARL})_{MBB}$	46.877	52.378	4.8103
Desired ARL		20		

Note: "Infinite" indicates some control limits is so large that no testing observation value goes beyond it. "---"implies the value can not be calculated accurately.

As shown in Table 1., PCA-based method can't obtain reliable charts for T^2. In addition, compared to the desired ARL , $\frac{1}{\alpha} = \frac{1}{0.05} = 20$, when b and k is properly selected , MBB-based $E(\mathrm{ARL})$ are expected to approach it, and the and $Std(\mathrm{ARL})$ be reasonable, while PCA-based method can't achieve this goal.

3.2 For FBR real data

In this example, 8 measured wall temperature variables of an industrial FBR were collected. The wall temperature monitoring plays an important role on monitoring the process of chunk formation on the wall of

Authorized licensed use limited to: UNIVERSITY OF NEW MEXICO. Downloaded on October 12,2026 at 16:31:59 UTC from IEEE Xplore. Restrictions apply.

the FBR. Autocorrelation test and normality test of the observations were completed, showing each of the variables is autocorrelated and does not follow normal distribution(test results are not shown here). The training observation size is 1000, and 50 observations under the same normal operations are selected to evaluate the 95% control limits by empirical false alarm rate. The results show that there are 8 false alarms in the 50 testing observations for the MBB Q chart , which has the same false alarm rate as the PCA-based chart. Since the noise level is high and strong autocorrelation exits for the industrial real data, some data-pretreatment should be done to improve the accuracy of MBB-based control limits . Other experiment results are not shown here and further studies are undergoing.

3.3 Results analysis

Sample size N : Generally speaking, for the sample of small size, following the same underlying distribution, MBB control charts are superior to PCA-based charts and the MBB-based method can also be applied to sample of large size.

Resample times B : large B can make the control limits more steady when other parameters are properly determined, because it reduce the influence of random resampling.

Block size b and block number k : in general, the larger the bk , the smaller the control limit ,in addition both $E(\mathrm{ARL})$ and $Std(\mathrm{ARL})$ is relatively small. This is caused by the data smoothing. And there should be interaction between b and k . Factor analysis or some optimization methods can be adopted to get the proper b and k for observations following different kind of distribution.

4 Conclusions

A method combining PCA with MBB for constructing MBB control chart for dependent multivariate data was presented in this paper. The MBB-based control limits are superior to the traditional PCA-based method under certain conditions, such as data don't satisfy IID distribution, and some crucial factors which has impact on MBB control chart were analyzed and the advantages of the MBB-based method were shown. As long as proper parameters were selected, the proposed approach is an available useful way to deal with some problem that tradition parametric method can't solve, such as the sample size is limited, underlying sampling distribution is unknown etc. However, this method can't be applied to highly dependent data, and the moving block size b and its number k is not easy to determine and need further extensive studies.

References

[1] B. Efron, "Bootstrap methods: another look at the Jackknife", *The annals of statistics*, vol. 7, No.1, 1-26,1979.

[2] J. H. Chen, and C. M. Liao et al, "Principal component analysis based control charts with memory effect for process monitoring", *Ind. Eng. Chem. Res.*, vol.40, No.6, 1516-1527, 2001.

[3] L. A. Jones, and W. H. Woodall, "The performance of bootstrap control charts", *Journal of quality technology*, Vol.30, No.4, 362-375, 1998.

[4] N. Choomrit, "Application criteria for bootstrap-based control chart", *Ph.D Dissertation*, Clemson University, 2001.

[5] R. Y. Liu, "Control charts for multivariate processes", *Journal of the American statistical association*, Vol.90, 1380-1387, 1995.

[6] R. Y. Liu, and J. Tang, "Control charts for dependent and independent measurements based on bootstrap methods", *Journal of the American statistical association*, Vol.91, 1694-1670, 1996.

[7] S. Wold, "Exponentially weighted moving principal components analysis projections to latent structure", *Chemometrics and intelligent laboratory systems*, 23, 149-161,1994.

[8] T. Seppala, H. Moskowitz, R. Plante and J. Tang, "Statistical process control via the subgroup bootstrap", *Journal of quality*, Vol. 27, No. 2, 139-153, 1995.

Authorized licensed use limited to: UNIVERSITY OF NEW MEXICO. Downloaded on October 12,2026 at 16:31:59 UTC from IEEE Xplore. Restrictions apply.